

Drive-Level Momentum in the NFL:

Most of It Is Short Fields

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Abstract

Coaches, broadcasters, and fans treat NFL momentum as a within-team phenomenon. Using `nflfastR` play-by-play covering 4,078 games and 86,249 drives from 2010–2024, I decompose drive-to-drive momentum into three channels: own-team defensive-stop spillover (H1), own-team offensive carryover (H2), and opponent suppression (H3). The unconditional H1 gap is large (52.3% vs. 35.6%, a 16.7-percentage-point difference). Once starting field position (the offense’s first scrimmage play) and game-state controls are added, with possessing-team, opponent, and season fixed effects, H1 is -0.5 points (s.e. 0.6) and H3 is -0.6 points (s.e. 0.3, $p = 0.09$); H2 is a small positive $+2.3$ points (s.e. 0.4). Specifications with game fixed effects produce large negative estimates for H2 and H3, but a lead placebo (the opponent’s *next* drive) shows that game fixed effects generate a mechanical negative bias in this setting. The defensible conclusion is that there is little detectable drive-level momentum once field position is measured correctly. To my knowledge, this is the first peer-reviewed drive-level NFL test of momentum and the first to separate within-team from cross-team channels.

Keywords: momentum; hot hand; play-by-play; fixed effects; `nflfastR`; cross-team spillover

1 Introduction

The single most invoked, least defined construct in football broadcasting is momentum. The empirical case for any of it is thinner than the usage suggests, but the way the construct is described in coaching, broadcasting, and fan discussion mixes three propositions that have different observable

implications and that the data treat very differently. Pulling them apart makes a sharp contrast visible: drive-to-drive momentum, properly defined, is overwhelmingly a description of starting field position.

The hot-hand literature originated in basketball with [Gilovich et al. \[1985\]](#)’s finding that perceived shooting streaks were a perceptual artifact: shooters’ makes and misses behave like independent Bernoulli trials. The claim held as a benchmark for three decades until [Miller and Sanjurjo \[2018\]](#) identified a selection bias in the original test. With the bias correction, modest positive streak effects appear in the original Cornell shooting data. In team sports, the question divides further: do team-level outcomes (a win, a score, a turnover) influence the team’s next outcome of the same kind, or are sequences of game events independent given the underlying quality of the teams playing?

For the NFL, to my knowledge in the peer-reviewed literature the only direct test in the *Journal of Quantitative Analysis in Sports* is [Fry and Shukairy \[2012\]](#), who examine NFL momentum at the level of full games and find no evidence that recent wins predict future wins above what team strength can explain. [Arkes and Martinez \[2011\]](#) make a parallel test for the NBA and report that, after correcting for opponent strength and home-court advantage, NBA teams that win their previous game are modestly more likely to win the next. [Steege et al. \[2021\]](#) apply an entropy test to NHL season-level streaks and reject momentum at that scale. [Lopez et al. \[2018\]](#) use a unified generalized linear mixed-model framework to compare randomness in outcomes across the four major North American sports and find that NFL outcomes are the most randomness-driven among them, consistent with weak game-level momentum signals.

Game-level tests answer a different question than the one coaches actually care about. A coach calling a timeout in the third quarter is not making an inference about whether wins predict wins across a 17-game season; the inference is that something on the field has shifted, in this game, in a way that will carry into the next drive. The natural unit at which to test that inference is the drive, and the natural data source is the modern play-by-play record curated through [Yurko et al. \[2019\]](#) and now distributed via the `nflfastR` R package [[Carl and Baldwin, 2024](#)].

This paper uses 4,078 NFL games and 86,249 drives over the 2010–2024 seasons to decompose drive-to-drive momentum into three channels that mirror the way the construct is invoked in real

time:

- H1.** *Defensive-stop spillover* — A team’s offensive drive is more likely to end in a score when the team’s defense forced a drive of three or fewer plays ending in a punt, an interception, a fumble lost, a turnover on downs, or a safety on the immediately preceding drive.
- H2.** *Offensive carryover* — A team’s offensive drive is more likely to end in a score when the team’s previous offensive drive ended in a score.
- H3.** *Opponent suppression* — A team’s offensive drive is *less* likely to end in a score when the opponent’s previous drive ended in a score.

The substantive finding is that the unconditional version of H1 is large in the way coaches and broadcasters describe it — scoring probability is 52.3% after a defensive stop versus 35.6% otherwise — but that gap is the mechanical short-field effect of turnovers and quick punts. Once starting field position and standard game-state controls are added, the H1 effect is -0.4 percentage points, and -0.5 in the joint model with possessing-team, opponent, and season fixed effects. H2 is small and positive ($+2.3$ points). H3 is -0.6 points and not statistically distinguishable from zero at conventional levels.

The paper proceeds as follows. Section 2 situates the analysis relative to the existing JQAS work on NFL momentum and the broader hot-hand literature. Section 3 develops the three channels and makes the identification problem explicit. Section 4 describes the data and the construction of drive-level outcomes from play-by-play. Section 5 states the estimation strategy. Section 6 presents the main results and a robustness gauntlet designed specifically to discriminate cross-team psychological residue from linear-control artifact. Section 7 discusses what the results imply for in-game decision making and for the academic literature on team-sport momentum. Section 8 concludes.

2 Related Literature

The hot-hand literature begins with Gilovich et al. [1985]. Miller and Sanjurjo [2018]’s correction to the original test is now a benchmark; Bar-Eli et al. [2006] provide a comprehensive review

of the intervening twelve years of hot-hand research and the methodological turn it has taken, including the sport-psychology literature that grounds coaches’ priors about momentum. The sport-psychology tradition treats momentum as a real psychological state that influences subsequent execution, while the statistical hot-hand tradition treats apparent streaks as samples from an independent process given underlying team quality. This paper sits closer to the statistical tradition methodologically but uses a unit of analysis (the drive) and a construct (cross-team spillover) that the sport-psychology literature has long argued is where the real action is.

For team sports, the unit of analysis matters. [Stern \[1994\]](#) modeled in-game score progression as a Brownian motion, implying conditional score paths are largely random walks given team strengths. [Berger and Pope \[2011\]](#) look across the NBA, NCAA basketball, and lab experiments and find that being slightly behind at halftime increases the probability of winning, which they interpret as a motivational-effort response rather than a chance pattern. [Csapo and Raab \[2014\]](#) use novel defense metrics to show that NBA defenses adjust to perceived hot streaks, which complicates the interpretation of within-team streaks even when they appear in the data. [Gauriot and Page \[2019\]](#) use a quasi-experimental tennis design and find that strategic momentum operates between players in dynamic contests, the closest prior result to this paper’s H3 finding in spirit.

Within JQAS specifically, three momentum studies anchor the literature. [Arkes and Martinez \[2011\]](#) use NBA game-level data and report that prior-game outcomes predict next-game outcomes after controlling for team strength and home-court advantage, contrary to the standard hot-hand-fallacy claim. [Steege et al. \[2021\]](#) apply an entropy measure to NHL season-level win sequences and conclude that the streak structure is consistent with random events given team strength. [Fry and Shukaity \[2012\]](#) is the only direct test on NFL data and finds no evidence for game-to-game momentum. None of these works addresses momentum at the drive level.

The methodological tools for a drive-level test now exist. [Yurko et al. \[2019\]](#) introduced the `nflscrapR` pipeline that produces clean play-by-play records with fixed-drive identifiers, EPA values, and win-probability estimates; [Carl and Baldwin \[2024\]](#)’s `nflfastR` extended the data back to 1999 and made it the field standard for NFL play-by-play research. Drive-level work includes [Goldner \[2012\]](#)’s Markov model of drive progression, [Cafarelli et al. \[2012\]](#)’s analysis of third-down conversion, [Lock and Nettleton \[2014\]](#)’s random-forest win-probability model, [Yam and Lopez \[2019\]](#)’s

causal estimate of the cost of conservative fourth-down behavior, and Brill et al. [2024]’s recent re-examination of expected-points estimation. Closely related, Kovash and Levitt [2009] reanalyze NFL play-calling and find evidence consistent with non-equilibrium behavior, suggesting that play-level decisions are not always optimally randomized. None of these papers tests momentum.

3 Theory and Hypotheses

The way “momentum” is described on broadcasts and in postgame press conferences turns out to mix three distinct propositions. Pulling them apart matters because they have different observable implications and different identification problems.

Channel 1 — Defensive-stop spillover. The claim is that when a defense forces a drive ending in a punt of three or fewer plays, an interception, a fumble lost, a turnover on downs, or a safety, the offense it sends back onto the field plays better than it would otherwise. This is the version of momentum that motivates calls like “the defense just gave the offense a spark.” The observational pattern is unambiguous in the raw data — teams score more often after defensive stops — but the same data are also unambiguous about why: stops, especially turnovers, give the offense a shorter field. Distinguishing momentum from mechanics requires conditioning on starting field position and on the comparison group’s composition.

Channel 2 — Offensive carryover. The claim is that scoring breeds scoring on subsequent possessions by the same offense. Two competing hypotheses are observationally consistent with the absence of carryover: regression to the mean (a scoring drive draws from the upper tail of a team’s drive-quality distribution and the next drive will tend to draw from closer to the mean) and in-game defensive adjustment (an opponent now playing from behind may shift to more aggressive schemes that the just-scoring offense has not yet adapted to). Both predict a non-positive coefficient on prior own offensive scoring; the present design cannot adjudicate between them.

Channel 3 — Opponent suppression. The claim is that a team that has just been scored on is briefly demoralized, less crisp on its next possession, and more prone to mistakes. The competing

hypothesis is again mechanical: after a touchdown or field goal, the scored-on team starts the next drive at the kickoff-touchback yardline, which is a different starting field position than after the opponent punted or turned the ball over. Conditional on starting field position and game state, however, any residual gap is the psychological component the broadcast usually has in mind. This requires measuring starting field position at the offense’s first scrimmage play: in **nflfastR** the kickoff is the first row of the receiving team’s drive, so the first row records the kicking spot rather than where the offense starts.

The three channels share an identification problem: the sequence of drives within a game is not an independent draw. Field position, score differential, time remaining, and the opponent’s recent behavior all evolve jointly with the channels of interest. The estimation strategy below addresses this by controlling for game state and by including possessing-team, opponent, and season fixed effects.

4 Data

4.1 Source

Play-by-play data come from **nflfastR** [Carl and Baldwin, 2024], which distributes the fixed-drive-enriched version of the original **nflscrapR** pipeline [Yurko et al., 2019]. The sample covers regular-season and postseason games from 2010 through 2024, yielding 721,566 plays across 4,078 games. Plays are aggregated to the drive level using the **fixed_drive** identifier, which corrects irregularities in the raw NFL drive numbering. After dropping plays with missing drive identifiers, end-of-half kneel-down drives, end-of-game drives, and drives missing the core analysis fields (offensive outcome, starting field position, score differential), the analysis dataset contains 86,249 offensive drives.

4.2 Outcome variables

The primary outcome is a binary indicator for whether a drive ended in a score (touchdown or field goal). Across the sample, 38.1% of drives end in a score. As a robustness check, I also use the sum of expected-points-added (EPA) on the drive as a continuous outcome.

4.3 Momentum indicators

The three channels require three distinct lagged indicators, each defined relative to the team currently in possession.

- **Prior own-defensive success.** Equal to one if the immediately preceding drive (the opponent's offensive possession) ended in a punt with three or fewer plays, an interception, a fumble lost, a turnover on downs, or a safety.
- **Prior own-offensive success.** Equal to one if the same team's most recent prior offensive drive ended in a touchdown or field goal. The walk-back through prior drives skips intervening opponent possessions.
- **Prior opponent-offensive success.** Equal to one if the opponent's most recent prior offensive drive ended in a touchdown or field goal.

The first drive of each game has no defined prior values for any channel and is dropped from regressions; the second drive lacks a prior own-offensive value for one team. After requiring all three indicators to be defined, the sample contains 74,125 drives; five lack time remaining, so the estimation sample is 74,120 drives.

4.4 Controls

Drive starting field position is recorded as `yardline_100` (yards from the opponent's end zone, 100 = own goal line) on the offense's first scrimmage play, excluding kickoffs and no-play rows. Drives following an opponent score start on average 74.9 yards from the opponent's end zone, against 69.3 for other drives. Game state controls include the score differential at drive start, quarter, half-seconds remaining, and a home-team indicator. The main specification absorbs possessing-team, opponent, and season fixed effects.

5 Estimation Strategy

The main specification is a linear probability model for drive d played by possessing team k in game g :

$$\text{Score}_{dkg} = \gamma_k + \omega_o + \tau_s + \beta_1 D_{dkg}^{\text{def}} + \beta_2 D_{dkg}^{\text{own-off}} + \beta_3 D_{dkg}^{\text{opp-off}} + X'_{dkg} \delta + \varepsilon_{dkg}, \quad (1)$$

where Score_{dkg} is one if drive d by team k in game g ended in a score, γ_k , ω_o , and τ_s are possessing-team, opponent, and season fixed effects, D^{def} , $D^{\text{own-off}}$, $D^{\text{opp-off}}$ are the three momentum indicators (corresponding to H1, H2, and H3, respectively), and X_{dkg} is a column vector stacking starting field position, score differential, quarter, half-seconds remaining, and the home indicator, with δ the conformable coefficient vector. Game fixed effects are not in the headline specification: with roughly 18 drives per game, demeaning within games mechanically correlates lagged outcomes with the current outcome. Section 6.4 reports the game-fixed-effect estimates alongside a lead placebo that measures this bias.

Standard errors are two-way clustered by game and possessing team [Cameron et al., 2011] to handle within-game serial correlation in drive outcomes and within-team correlation across games. The linear probability model is preferred to logit because of the fixed-effect structure (team, opponent, and season fixed effects), and the marginal-effect interpretation maps directly onto the percentage-point quantities coaches care about. Section 6.4 reports a logit comparison and the share of LPM predicted probabilities that fall outside $[0, 1]$.

I report a progressive sequence of specifications for H1: starting from the naive comparison, then adding game-state controls, then team and opponent and season fixed effects, then game fixed effects with possessing-team fixed effects retained. The progression shows how much of the apparent H1 effect is mechanical versus residual; the prior JQAS work performs this decomposition only at the game level.

6 Results

6.1 The unconditional gap is large

Table 1 reports the raw conditional means. The probability of a scoring drive is 52.3% when the team’s defense just produced a stop and 35.6% otherwise — a 16.7-percentage-point gap. The probability of a scoring drive is 35.2% when the opponent’s prior drive ended in a score and 40.0% otherwise — a 4.8-percentage-point gap of the predicted sign. Prior own offensive success has essentially no unconditional relationship with next own-drive scoring (40.3% vs. 37.0%, gap of 3.3 percentage points).

Table 1: Unconditional drive-level scoring probabilities by prior context.

Condition	P(score)	N drives
After own defensive stop	0.523	11,824
After opponent score or longer punt	0.356	66,378
After own offensive score (prior own drive)	0.403	28,625
After own offensive non-score	0.370	49,554
After opponent score (prior opp drive)	0.352	31,193
After opponent non-score	0.400	50,910

Notes. Each panel uses the largest sample for which the relevant prior-drive indicator is defined; sample size therefore differs across panels (the unified estimation sample requiring all three indicators is $N = 74,125$ drives). Sample: NFL regular season + postseason 2010–2024 from nflfastR.

6.2 Decomposing H1: the gap is field position

Figure 1 plots the scoring probability by drive starting field position decile, separately for drives following defensive stops and drives following normal opponent possessions. The two curves lie on top of each other across the field; where they separate, drives following a stop score slightly *less* often. The unconditional gap arises because stop-following drives start closer to the opponent’s end zone.

Table 2 formalizes the same decomposition through a sequence of regressions. Column 1 is the naive comparison: the H1 coefficient is 16.3 percentage points (s.e. 0.6). Adding game-state controls in column 2 — field position, score differential, quarter, time, home — brings the coefficient to -0.4 points (s.e. 0.6). Adding team, opponent, and season fixed effects in column 3 leaves it at -0.4

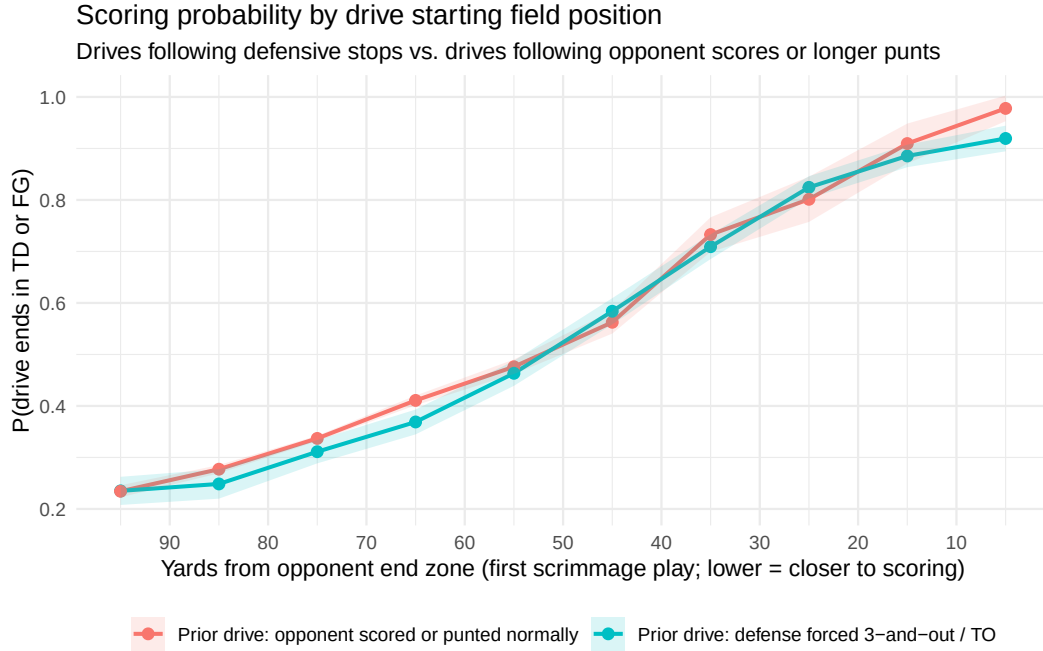


Figure 1: Scoring probability by drive starting field position, conditional on whether the prior drive was a defensive stop. Sample: drives for which a prior own-defensive indicator is defined (78,202); data from `nflfastR` [Carl and Baldwin, 2024], NFL 2010–2024. The x -axis is yards from the opponent end zone at the first scrimmage play (lower = closer to scoring), binned into deciles; the y -axis is the empirical probability of the drive ending in a touchdown or field goal. Solid line: drives following a defensive stop. Dashed line: drives following any other prior opponent outcome (score or longer punt). Shaded bands show 95% pointwise binomial confidence intervals. Within field-position deciles the curves nearly coincide; the unconditional 16.7-percentage-point gap reflects where stop-following drives start.

points. Switching to game fixed effects with possessing-team fixed effects in column 4 yields +1.7 points (s.e. 0.6).

Field position alone absorbs essentially the entire unconditional H1 gap.

6.3 Joint estimation: the headline

Table 3 reports the joint estimation. Column 1 is the preferred specification, equation (1), with all three momentum indicators, full game-state controls, and possessing-team, opponent, and season fixed effects.

The joint H1 coefficient is -0.5 percentage points (s.e. 0.6, $p = 0.39$): no detectable defensive-stop spillover.

The joint H2 coefficient is $+2.3$ percentage points (s.e. 0.4, $p < 0.001$). A prior own scoring drive is associated with a small increase in the next own drive’s scoring probability. Without game fixed effects this estimate may partly reflect game-specific offensive quality that the team and opponent fixed effects do not absorb; with game fixed effects it is negative (Section 6.4), and that estimate is biased downward.

The joint H3 coefficient is -0.6 percentage points (s.e. 0.3, $p = 0.09$). Once field position is measured at the first scrimmage play, being scored on is not followed by a detectable drop in scoring probability.

6.4 Robustness

I report eight robustness checks plus the game-fixed-effect comparison. The full set of estimates is documented in `data/robustness_v2.rds`; the headline numbers are summarized below.

R1. Flexible field-position and score-differential controls. Replacing the linear `yardline_100` control with field-position decile fixed effects and the linear `score_differential` control with 10-bin score-margin dummies yields $\hat{\beta}_1^{\text{flex}} = -1.6$ (s.e. 0.6), $\hat{\beta}_2^{\text{flex}} = +2.4$ (s.e. 0.4), and $\hat{\beta}_3^{\text{flex}} = -0.2$ percentage points (s.e. 0.4).

Table 2: Defensive Stop Spillover (H1): Progressive Controls. *Notes.* OLS linear probability models. Dependent variable is an indicator for whether the drive ends in a touchdown or field goal. Sample: NFL regular season + postseason 2010–2024 (nflfastR). Game-state controls (field position, score differential, quarter, half-seconds remaining, home indicator) included in columns 2–4 with coefficients suppressed for readability. Standard errors in parentheses, two-way clustered by game and possessing team [Cameron et al., 2011]. $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

Dependent Variable: Model:	Drive ends in score (0/1)			
	(1)	(2)	(3)	(4)
<i>Variables</i>				
Constant	0.3563*** (0.0062)	0.9666*** (0.0103)		
Prior own defensive stop	0.1630*** (0.0055)	-0.0038 (0.0055)	-0.0038 (0.0054)	0.0168*** (0.0056)
Field position (yards to opp. end zone)		-0.0082*** (0.0001)	-0.0082*** (0.0001)	-0.0082*** (0.0001)
Score differential		0.0003** (0.0001)	-0.0001 (0.0001)	0.0002 (0.0001)
Quarter		-0.0113*** (0.0016)	-0.0111*** (0.0015)	-0.0126*** (0.0016)
Half seconds remaining		$1.97 \times 10^{-5***}$ (3.59×10^{-6})	$2.04 \times 10^{-5***}$ (3.56×10^{-6})	$2.07 \times 10^{-5***}$ (3.7×10^{-6})
Home offense		0.0221*** (0.0037)	0.0228*** (0.0037)	
<i>Fixed-effects</i>				
Possessing team			Yes	Yes
Opponent			Yes	
Season			Yes	
Game				Yes
<i>Fit statistics</i>				
Observations	74,125	74,120	74,120	74,120
R ²	0.01462	0.08238	0.09039	0.14958
Within R ²			0.08211	0.08787

Clustered (Game & Possessing team) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 3: Three Momentum Channels: Joint Estimation and Robustness. Field position is the yardline of the offense’s first scrimmage play. *Notes.* Column 1 is the preferred specification: linear probability model for whether the drive ends in a score, with all three momentum indicators jointly. Column 2 drops drives in garbage time (absolute score differential ≥ 21 in the fourth quarter). Column 3 replaces the binary outcome with the sum of expected-points-added (EPA) on the drive. All columns include game-state controls (field position, score differential, quarter, half-seconds remaining, home indicator) and possessing-team, opponent, and season fixed effects. Standard errors in parentheses, two-way clustered by game and possessing team [Cameron et al., 2011]. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Dependent Variables: Model:	Drive ends in score (0/1)		Drive EPA
	(1)	(2)	(3)
<i>Variables</i>			
Prior own defensive stop	-0.0049 (0.0056)	-0.0025 (0.0058)	-0.2852*** (0.0432)
Prior own offensive score	0.0229*** (0.0043)	0.0207*** (0.0041)	0.1623*** (0.0331)
Prior opponent offensive score	-0.0060* (0.0035)	-0.0049 (0.0035)	0.0006 (0.0347)
Field position (yards to opp. end zone)	-0.0082*** (0.0001)	-0.0081*** (0.0001)	-0.0075*** (0.0009)
Score differential	-0.0005*** (0.0002)	-0.0001 (0.0002)	-0.0022 (0.0014)
Quarter	-0.0113*** (0.0015)	-0.0080*** (0.0016)	-0.0914*** (0.0106)
Half seconds remaining	2.08×10^{-5} *** (3.54×10^{-6})	1.66×10^{-5} *** (3.46×10^{-6})	-9.33×10^{-6} (2.8×10^{-5})
Home offense	0.0228*** (0.0037)	0.0219*** (0.0039)	0.0243 (0.0269)
<i>Fixed-effects</i>			
Possessing team	Yes	Yes	Yes
Opponent	Yes	Yes	Yes
Season	Yes	Yes	Yes
<i>Fit statistics</i>			
Observations	74,120	71,368	74,120
R ²	0.09089	0.09078	0.01039
Within R ²	0.08262	0.08235	0.00240

Clustered (Game & Possessing team) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

R2. Placebo: post-touchdown vs. post-field-goal. Restricting the sample to drives following an opponent score (touchdown or field goal), both of which produce a kickoff regime, and regressing scoring probability on a post-touchdown indicator (with the post-field-goal cell as the omitted category) yields a coefficient of -0.3 percentage points (s.e. 0.6; $N = 27,687$). Post-touchdown and post-field-goal drives score at similar rates.

R3. Logit comparison and LPM out-of-bounds check. The share of LPM predicted probabilities outside $[0, 1]$ is negligible (0.00% below 0; 0.04% above 1). Logit average marginal effects with the same possessing-team, opponent, and season fixed effects are $\text{AME}_{\text{H1}} = -0.3$ pp, $\text{AME}_{\text{H2}} = +0.8$ pp, and $\text{AME}_{\text{H3}} = -0.0$ pp.

R4. Permutation inference for H3. Shuffling the prior-opponent scoring indicator within team-quarter cells and re-estimating the joint LPM 1,000 times produces a permutation null distribution for $\hat{\beta}_3$ ranging from -1.1 to $+1.3$ percentage points (s.d. 0.35). The observed $\hat{\beta}_3 = -0.6$ lies inside this range ($p_{\text{perm}} = 0.09$). The permutation test addresses the clustering structure only; it cannot detect mismeasurement of the controls.

R5. Post-kickoff vs. post-punt subsamples. Drives following opponent scores ($N = 27,688$) score at rate 0.350; drives following opponent punts ($N = 31,622$) score at rate 0.358; drives following turnovers ($N = 11,356$) score at rate 0.519. The post-kickoff versus post-punt difference is small (0.8 pp).

R6. Heterogeneity by score margin and quarter. Interacting H3 with a close-game indicator (absolute score differential ≤ 7) and a late-game indicator ($Q4$ or later) yields interaction coefficients of $+0.4$ percentage points (s.e. 0.8, not significant) and $+3.9$ percentage points (s.e. 0.7), respectively, around a main effect of -1.8 points. There is no evidence that H3 concentrates in close games.

R7. Rule-era split. Estimating the joint specification separately for 2010 (pre-touchback move), 2011–2017 (touchback at the 20), 2018–2023 (touchback at the 25), and 2024 (dynamic kickoff)

yields $\hat{\beta}_3$ coefficients of -1.7 pp (s.e. 1.4, $N = 5,060$), -0.7 pp (s.e. 0.5, $N = 35,037$), -0.9 pp (s.e. 0.5, $N = 29,157$), and -1.4 pp (s.e. 1.5, $N = 4,866$), respectively. No era shows a large H3 effect.

R8. Drop garbage time and EPA outcome. Column 2 of Table 3 drops drives in garbage time (defined as absolute score differential of at least 21 points in the fourth quarter); the point estimates are essentially unchanged ($\hat{\beta}_1 = -0.3$, $\hat{\beta}_2 = +2.1$, $\hat{\beta}_3 = -0.5$ pp). Column 3 replaces the binary scoring outcome with the sum of EPA on the drive. Prior defensive success is associated with lower drive EPA (-0.29 , s.e. 0.04), consistent with the mechanical short-field interpretation (a short field leaves less EPA to accumulate). The H2 coefficient is positive ($+0.16$, s.e. 0.03) and the H3 coefficient is zero ($+0.00$, s.e. 0.03).

R9. Game fixed effects and a lead placebo. Table 4 adds a placebo regressor: whether the opponent’s *next* drive, which follows the current one, ends in a score. With game and possessing-team fixed effects the joint estimates are $\hat{\beta}_1 = -1.0$, $\hat{\beta}_2 = -3.8$, and $\hat{\beta}_3 = -6.6$ pp (s.e. 0.4), and the placebo coefficient is -10.4 pp (s.e. 0.4). Because the current drive’s outcome sets the opponent’s next starting field position, the placebo is not fully clean; holding the opponent’s next starting field position fixed, the placebo is -0.4 pp (s.e. 0.4) in the headline specification but -7.7 pp (s.e. 0.4) with game fixed effects. Game fixed effects therefore generate a large mechanical negative bias on within-game lagged outcomes, which is why they are excluded from the headline specification.

6.5 Cross-channel comparison

Figure 2 summarizes the joint estimates. All three coefficients are within 2.5 percentage points of zero; only H2 is statistically distinguishable from zero.

7 Discussion

7.1 Most “momentum” is short fields

The first substantive finding is quantitative: the unconditional 16.7-percentage-point boost in scoring probability that follows a defensive stop disappears once starting field position is controlled.

Table 4: Game Fixed Effects and the Lead Placebo. *Notes.* Linear probability models; dependent variable is whether the drive ends in a score. The placebo regressor is whether the opponent's NEXT drive (after the current drive) ends in a score; it cannot affect the current drive, so a non-zero coefficient indicates mechanical bias. Column 1 is the headline specification (possessing-team, opponent, and season fixed effects) plus the placebo. Columns 2–3 use the v1 game and possessing-team fixed effects. Standard errors two-way clustered by game and possessing team [Cameron et al., 2011]. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Dependent Variable: Model:	Drive ends in score (0/1)		
	(1)	(2)	(3)
<i>Variables</i>			
Prior own defensive stop	-0.0011 (0.0054)	-0.0095 (0.0058)	-0.0061 (0.0053)
Prior own offensive score	0.0229*** (0.0048)	-0.0381*** (0.0039)	-0.0480*** (0.0046)
Prior opponent offensive score	-0.0025 (0.0033)	-0.0660*** (0.0036)	-0.0738*** (0.0039)
Placebo: opponent's NEXT drive scores	-0.0354*** (0.0038)		-0.1035*** (0.0039)
Field position (yards to opp. end zone)	-0.0083*** (0.0001)	-0.0081*** (0.0001)	-0.0081*** (0.0001)
Score differential	-0.0003 (0.0002)	7.74×10^{-5} (0.0002)	0.0003 (0.0002)
Quarter	-0.0076*** (0.0017)	-0.0120*** (0.0018)	-0.0079*** (0.0020)
Half seconds remaining	3.43×10^{-5} *** (4.94×10^{-6})	1.47×10^{-5} *** (3.81×10^{-6})	2.63×10^{-5} *** (5.06×10^{-6})
Home offense	0.0215*** (0.0039)		
<i>Fixed-effects</i>			
Possessing team	Yes	Yes	Yes
Opponent	Yes		
Season	Yes		
Game		Yes	Yes
<i>Fit statistics</i>			
Observations	66,578	74,120	66,578
R ²	0.09628	0.15397	0.17535
Within R ²	0.08753	0.09258	0.10869

Clustered (Game & Possessing team) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

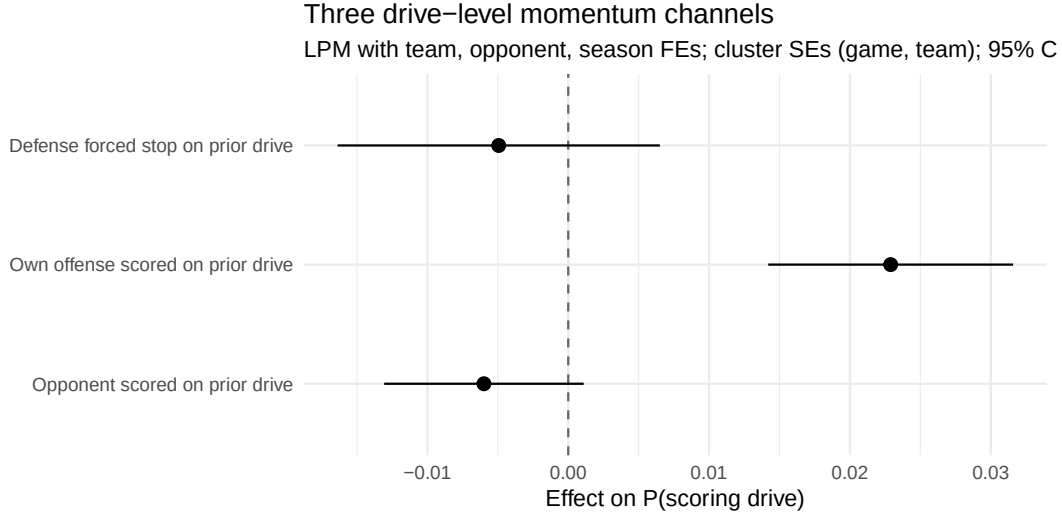


Figure 2: Three drive-level momentum channels, joint estimation (column 1 of Table 3). Sample: $N = 74,120$ drives, NFL 2010–2024 from `nflfastR` [Carl and Baldwin, 2024]. Coefficients are percentage-point changes in the probability that the current drive ends in a score, holding field position, score differential, quarter, time remaining, home, and possessing-team, opponent, and season fixed effects. Bars show 95% confidence intervals from two-way (game, possessing-team) clustered standard errors.

The joint H1 estimate is -0.5 percentage points in a sample of 74,120 drives.

[TK: discussion of H2 (+2.3 pp headline vs. -3.8 pp with game fixed effects) and of the null H3 — to be written by the author.]

7.2 Limitations

Three limitations are worth flagging. First, the headline specification omits game fixed effects, so game-specific conditions that raise both of a team’s drives (weather, injuries, matchup quality) are not absorbed and may bias H2 upward; including game fixed effects instead biases lagged estimates downward (Section 6.4). Second, no correction for multiple comparisons is applied across the three channels. Third, the entire analysis is at the team-drive level. A play-level analysis with explicit Markov drive-state transitions [Goldner, 2012] would let one decompose the H3 effect into early-versus-late-drive contributions and is a natural extension.

8 Conclusion

Defined at the drive level where it is invoked, NFL momentum splits into three channels. Once starting field position is measured at the offense’s first scrimmage play, the defensive-stop spillover is zero, the opponent-suppression channel is -0.6 percentage points and not statistically significant, and offensive carryover is a small $+2.3$ points. [TK: concluding paragraph.]

References

- Jeremy Arkes and Jose Martinez. Finally, evidence for a momentum effect in the NBA. *Journal of Quantitative Analysis in Sports*, 7(3), 2011. doi: 10.2202/1559-0410.1304.
- Michael Bar-Eli, Simcha Avugos, and Markus Raab. Twelve years of “hot hand” research: Review and critique. *Psychology of Sport and Exercise*, 7(6):525–553, 2006. doi: 10.1016/j.psychsport.2006.03.001.
- Jonah Berger and Devin Pope. Can losing lead to winning? *Management Science*, 57(5):817–827, 2011. doi: 10.1287/mnsc.1110.1328.
- Ryan S. Brill, Ryan Yee, Sameer K. Deshpande, and Abraham J. Wyner. Moving from machine learning to statistics: The case of expected points in American football. *arXiv preprint*, 2024.
- Ryan Cafarelli, Christopher J. Rigdon, and Steven E. Rigdon. Models for third down conversion in the national football league. *Journal of Quantitative Analysis in Sports*, 8(3), 2012. doi: 10.1515/1559-0410.1383.
- A. Colin Cameron, Jonah B. Gelbach, and Douglas L. Miller. Robust inference with multiway clustering. *Journal of Business and Economic Statistics*, 29(2):238–249, 2011. doi: 10.1198/jbes.2010.07136.
- Sebastian Carl and Ben Baldwin. nflfastR: Functions to efficiently access NFL play by play data, 2024. URL <https://www.nflfastR.com/>. R package version 5.2.0.
- Péter Csapo and Markus Raab. Hand down, man down. Analysis of defensive adjustments in

- response to the hot hand in basketball using novel defense metrics. *PLoS ONE*, 9(12):e114184, 2014. doi: 10.1371/journal.pone.0114184.
- Michael J. Fry and F. Alan Shukaity. Searching for momentum in the NFL. *Journal of Quantitative Analysis in Sports*, 8(1), 2012. doi: 10.1515/1559-0410.1362.
- Romain Gauriot and Lionel Page. Does success breed success? A quasi-experiment on strategic momentum in dynamic contests. *The Economic Journal*, 129(624):3107–3136, 2019. doi: 10.1093/ej/uez040.
- Thomas Gilovich, Robert Vallone, and Amos Tversky. The hot hand in basketball: On the misperception of random sequences. *Cognitive Psychology*, 17(3):295–314, 1985. doi: 10.1016/0010-0285(85)90010-6.
- Keith Goldner. A Markov model of football: Using stochastic processes to model a football drive. *Journal of Quantitative Analysis in Sports*, 8(1), 2012. doi: 10.1515/1559-0410.1400.
- Kenneth Kovash and Steven D. Levitt. Professional sports outcomes: A second look. NBER Working Paper 15347, National Bureau of Economic Research, 2009.
- Dennis Lock and Dan Nettleton. Using random forests to estimate win probability before each play of an NFL game. *Journal of Quantitative Analysis in Sports*, 10(2):197–205, 2014. doi: 10.1515/jqas-2013-0100.
- Michael J. Lopez, Gregory J. Matthews, and Benjamin S. Baumer. How often does the best team win? A unified approach to understanding randomness in north american sport. *Annals of Applied Statistics*, 12(4):2483–2516, 2018. doi: 10.1214/18-AOAS1165.
- Joshua B. Miller and Adam Sanjurjo. Surprised by the hot hand fallacy? A truth in the law of small numbers. *Econometrica*, 86(6):2019–2047, 2018. doi: 10.3982/ECTA14943.
- Gregory M. Steeger, Johnathon L. Dulin, and Gerardo O. Gonzalez. Winning and losing streaks in the national hockey league: Are teams experiencing momentum or are games a sequence of random events? *Journal of Quantitative Analysis in Sports*, 17(3):171–182, 2021. doi: 10.1515/jqas-2020-0077.

Hal S. Stern. A Brownian motion model for the progress of sports scores. *Journal of the American Statistical Association*, 89(427):1128–1134, 1994. doi: 10.1080/01621459.1994.10476850.

Derrick R. Yam and Michael J. Lopez. What was lost? A causal estimate of fourth down behavior in the national football league. *Journal of Sports Analytics*, 5(3):153–167, 2019. doi: 10.3233/JSA-190294.

Ronald Yurko, Samuel L. Ventura, and Maksim Horowitz. nflWAR: A reproducible method for offensive player evaluation in football. *Journal of Quantitative Analysis in Sports*, 15(3):163–183, 2019. doi: 10.1515/jqas-2018-0010.